

WEEK 5 – LECTURE 1

Artificial Intelligence Foundations

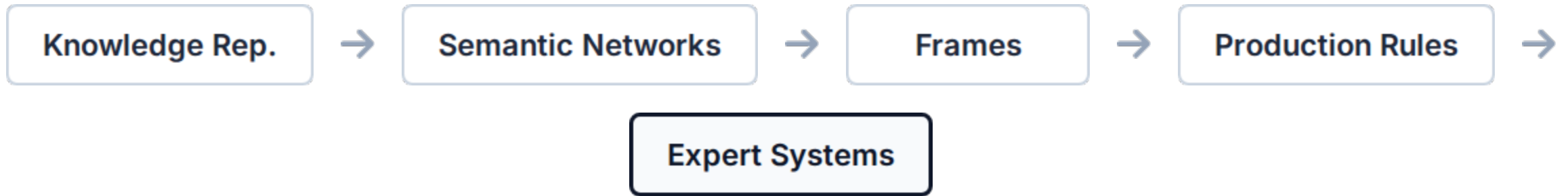
Propositional Logic & First-Order Predicate Logic

Formal Languages for Knowledge Representation

| CSE276 | Unit II



Quick Recap



Can AI represent knowledge using a universal, mathematical language?

Motivation

"The classroom lights are ON."

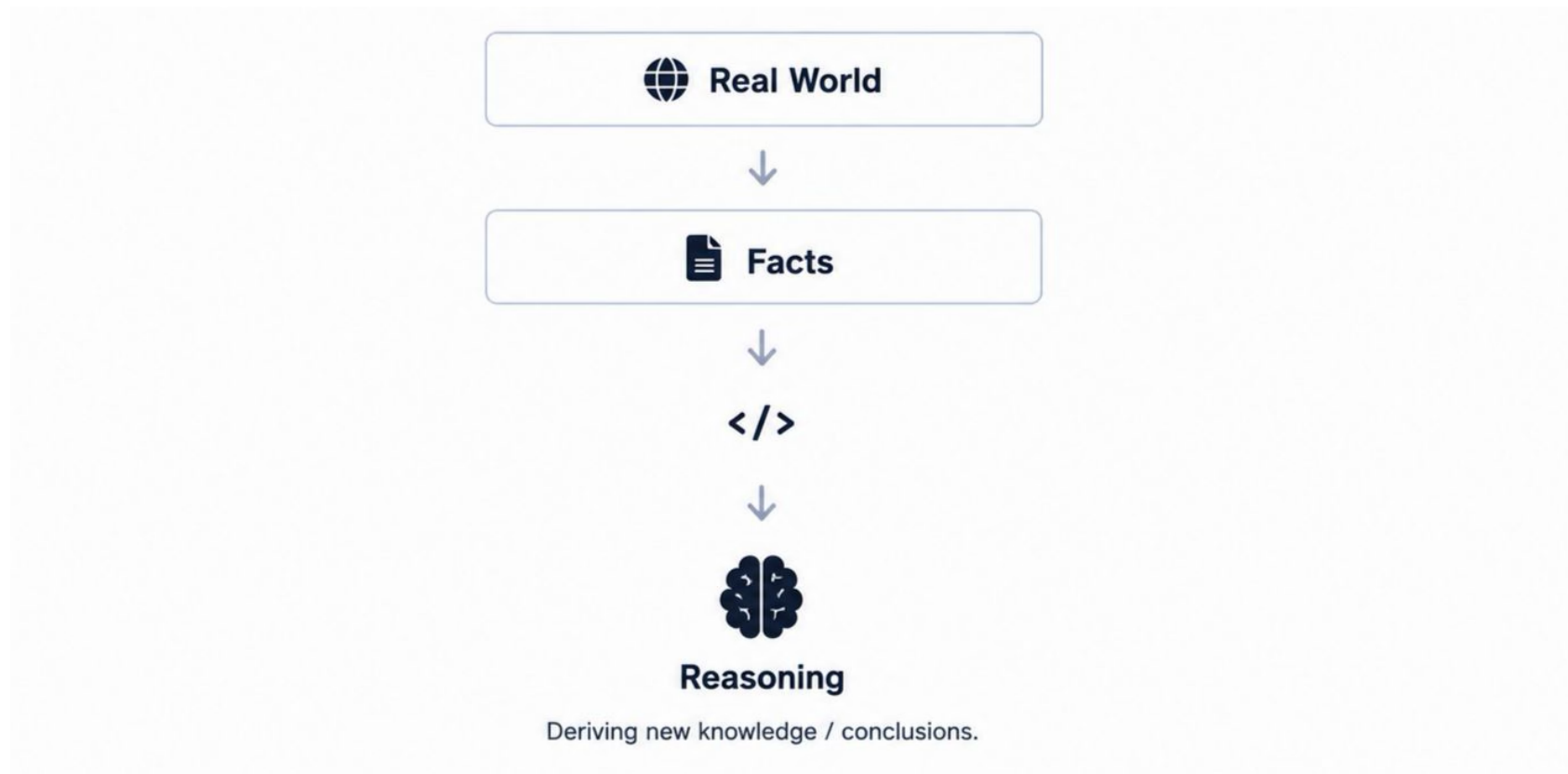
"Ashley is a student."

"Every student has an ID card."

How can AI represent these facts so that a computer can understand and reason about them?

What is Logic?

A formal language for representing knowledge and reasoning.



What is Propositional Logic?

A proposition is a statement that is either **True** or **False**.

Examples

- The sun rises in the east. **(T)**
- Delhi is the capital of India. **(T)**
- $5 > 10$ **(F)**

Not Propositions

What time is it?

Please close the door.

$x + y = 10$ (Depends on x , y)

Logical Connectives

Used to combine simple propositions into complex ones.

$\wedge$

AND

Conjunction

$\vee$

OR

Disjunction

$\neg$

NOT

Negation

$\rightarrow$

IMPLIES

Implication

"If attendance $\geq$ 75% **AND** exam passed $\rightarrow$ Eligible."

Truth Tables

P	Q	$P \wedge Q$ (AND)
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T T T

T F F

F T F

F F F

P	Q	$P \vee Q$ (OR)
---	---	-----------------

T T T

T F T

F T T

F F F

P	$\neg P$ (NOT)
---	----------------

T F

F T

Interactive Activity

P = "It is raining." (True)

Q = "I have an umbrella." (False)

Determine the result for:

1) $P \wedge Q$

False

True

Propositional Logic in Action

P = Attendance $\geq$ 75%

Q = Passed Exam

$P \wedge Q$ "Attendance is $\geq$ 75% AND passed exam."

$P \vee Q$ "Attendance is $\geq$ 75% OR passed exam."

$\neg P$ "Attendance is NOT $\geq$ 75%."

$P \rightarrow Q$ "IF attendance is $\geq$ 75%, THEN passed exam."

Limitations of Propositional Logic

How do we represent:
"Every student studies AI."

- In Propositional Logic, we would need a separate proposition for *every single student*:

P1 = "Alice studies AI"

P2 = "Bob studies AI"

P3 = "Charlie studies AI" ...

**This becomes impossible for large numbers of objects.
We need a more expressive language!**

First-Order Predicate Logic (FOPL)

Extends propositional logic by breaking down sentences into inner structures.



Objects

Things in the world (Rishi, AI, Delhi)



Properties

Attributes of objects (is a Student)



Relationships

Connections between objects (Studies)

Student (Karan) → "Karan is a student."

Studies (Karan, AI) → "Karan studies AI."

First Order Logic in AI

X is an integer

Subject

Predicate

Components of Predicate Logic

Objects (Constants)

Rishi, Alice, Delhi, Pizza

Predicates

Student(x), Likes(x, y)

Variables

x, y, z (stand for any object)

Simple Examples

Likes (Riya , Pizza)

LivesIn (Alice , Delhi)

Quantifiers

Allow us to express statements about collections of objects.



Universal Quantifier

"For All" or "Every"

$\forall x \text{ Student}(x) \rightarrow \text{Studies}(x)$

"Every student studies."



Existential Quantifier

"There Exists" or "Some"

$\exists x \text{ Student}(x) \wedge \text{Scored}(x, 100)$

"Some student scored 100."

University Scenario Examples



Student (Max)

Max is a student.



Studies (Max , AI)

Max studies AI.



Faculty (Rishi)

Rishi is faculty.



Teaches (Rishi , AI)

Rishi teaches AI.

Propositional vs Predicate Logic

Feature	Propositional Logic	First-Order Predicate Logic
Represents	Entire facts (True/False)	Objects, Properties, Relationships
Variables	P, Q (Whole statements)	x, y (Individual objects)
Expressiveness	Limited	Highly Expressive
Quantifiers ($\forall, \exists$)	No ❌	Yes ✅
Example	P = "All birds fly"	$\forall x \text{ Bird}(x) \rightarrow \text{CanFly}(x)$

AI Applications of Logic

Logical reasoning underpins many classic and modern AI systems.



Expert Systems



Robotics Planning



**Rule-Based
Chatbots**



**Knowledge
Graphs**



**Access Control
Systems**



Board Activity

Translate to Predicate Logic:

-  1. Every student has an ID card.
-  2. Some students play football.
-  3. Riya studies AI.
-  4. Every teacher teaches at least one subject.

Common Misconceptions



"Logic is the same as programming."

Correction: Logic represents facts; programming executes instructions.



"Predicate Logic completely replaces Propositional Logic."

Correction: Propositional logic is still used for simple, fast evaluations.



"Logical statements are always extremely complicated."

Correction: They are just structured ways to write everyday sentences.

Key Takeaways



Logic represents
knowledge formally.



Propositional Logic
handles True/False
facts.



Predicate Logic
handles objects and
relationships.



Predicate logic is more
expressive for AI.

Quick Quiz

Which logic is needed to express the relationship between specific objects (e.g., "Alice lives in Delhi")?

- ☐ A) Propositional Logic
- ☐ B) First-Order Predicate Logic
- ☐ C) Semantic Networks
- ☐ D) Boolean Algebra

Quick Quiz

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Preview: Next Lecture

Reasoning Under Uncertainty

What happens when facts are not 100% True or False?

 Probability Basics

 Bayes' Theorem

 Decision making under uncertainty